

KABUL INTL, Afghanistan

WMO: 409480

Lat: 34.5552N

Lon: 69.2142E

Elev: 5877

StdP: 11.83

Time Zone: 4.50 (AFG)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	19.7	-5.5	5.1	36.0	-1.8	6.2	38.1	15.1	38.5	12.7	36.9	6.6	310	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	95.3	58.3	93.5	57.8	91.6	57.5	67.5	80.7	66.1	79.2	64.5	78.0	9.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	111.2	74.9	61.2	100.7	74.4	59.0	93.1	74.9	35.8	81.2	34.5	79.5	33.0	78.4	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.5	17.6	13.2	99.6	7.2	1.5	8.1	100.7	3.9	101.5	-0.2	102.4	-5.4	103.5
			10.5	70.0	6.3	1.4	6.0	71.0	2.3	71.9	-1.2	72.7	-5.8	73.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	34.1	37.6	49.4	58.6	67.5	75.6	80.0	78.3	71.4	60.3	48.2	39.2
	DBStd	16.65	6.19	6.85	6.76	6.28	5.63	4.33	3.06	3.16	4.29	5.52	5.23	5.27
	HDD50	1372	492	349	92	7	0	0	0	0	0	1	94	337
	HDD65	3931	957	766	483	206	41	1	0	0	6	167	504	800
	CDD50	4466	0	3	75	265	542	769	929	878	643	321	40	1
	CDD65	1550	0	0	0	14	118	320	464	413	199	22	0	0
	CDH74	19185	0	0	13	252	1436	4033	6052	4868	2161	366	4	0
	CDH80	8531	0	0	0	43	427	1788	3117	2347	744	65	0	0
Wind	WSAvg	6.3	4.6	4.9	5.6	6.3	7.9	9.8	9.2	7.3	6.1	5.1	4.2	4.3
Precipitation	PrecAvg	15.3	1.7	2.8	2.6	2.0	1.3	0.6	0.6	0.9	0.6	0.6	0.8	0.8
	PrecMax	25.9	4.0	5.5	6.6	4.8	5.1	2.3	1.9	3.5	1.9	2.3	3.0	2.8
	PrecMin	4.9	0.2	0.2	0.2	0.3	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.1
	PrecStd	4.8	1.1	1.4	1.6	1.2	1.1	0.6	0.4	0.7	0.6	0.7	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	65.9	75.4	84.5	90.0	96.7	98.8	97.1	91.7	85.7	73.6	62.8
		MCWB	37.7	44.2	50.0	53.2	54.9	57.8	59.5	58.9	57.1	52.2	47.2	42.4
	2%	DB	52.1	57.3	71.2	80.4	87.7	93.5	96.5	94.9	89.4	82.0	68.3	57.4
		MCWB	36.6	41.3	48.1	52.7	54.3	56.6	59.5	58.3	56.1	51.9	44.8	39.7
	5%	DB	48.5	53.3	66.6	76.9	84.5	91.4	94.6	92.9	86.4	78.5	64.6	53.8
		MCWB	34.9	40.7	46.6	52.3	53.6	56.5	58.4	58.0	55.3	50.6	43.4	38.0
	10%	DB	44.9	48.6	62.8	73.2	82.2	89.2	92.8	90.0	84.5	74.9	60.9	50.4
		MCWB	33.9	38.4	45.5	51.2	52.9	55.8	58.1	57.7	54.9	49.4	42.5	36.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.7	46.5	52.4	57.5	58.8	66.2	69.5	68.9	66.1	59.7	50.1	44.8
		MCDB	47.9	55.7	67.0	74.6	77.3	80.0	84.0	81.3	79.5	73.1	62.3	59.0
	2%	WB	38.4	44.3	50.7	55.3	57.0	62.6	67.9	67.5	64.0	56.2	47.9	42.1
		MCDB	46.0	54.4	63.8	71.5	75.3	80.0	81.9	79.7	77.4	70.9	61.5	53.0
	5%	WB	36.7	42.7	49.2	54.0	55.8	59.3	66.3	66.3	62.2	53.5	46.0	40.0
		MCDB	44.3	49.8	61.2	70.2	74.9	82.1	80.5	78.4	75.5	69.3	58.9	49.3
	10%	WB	35.1	40.4	47.7	52.5	54.8	57.2	64.3	64.8	60.1	51.5	44.4	38.1
		MCDB	42.0	47.1	59.3	67.9	74.4	84.4	78.6	77.6	74.4	70.0	56.9	47.6
Mean Daily Temperature Range	5% DB	MDBR	19.4	17.9	20.5	22.8	25.6	25.7	25.5	25.2	26.5	27.2	23.9	22.4
		MCDBR	24.9	24.2	27.2	28.8	28.9	28.6	28.1	28.2	28.0	29.9	29.6	27.5
		MCWBR	13.2	12.0	10.7	9.8	8.8	9.7	10.7	11.2	10.7	11.2	13.1	13.8
	5% WB	MCDBR	19.5	20.1	21.7	23.7	24.7	25.6	24.2	22.9	23.6	24.8	24.0	23.5
		MCWBR	11.0	10.5	9.4	9.4	8.4	10.5	11.2	10.6	10.7	10.0	10.8	12.1
Clear-Sky Solar Irradiance	taub	0.284	0.327	0.377	0.410	0.415	0.430	0.440	0.411	0.357	0.336	0.306	0.285	
	taud	2.391	2.227	2.133	2.098	2.113	2.069	2.053	2.150	2.293	2.310	2.394	2.420	
	Ebn at Noon	294	289	282	278	277	271	268	274	285	283	283	287	
	Edh at Noon	29	38	46	50	50	52	53	47	39	35	29	27	
All-Sky Solar Radiation	RadAvg	939	1164	1549	1919	2320	2568	2533	2300	2025	1609	1110	909	
	RadStd	89	136	147	148	142	113	70	127	77	74	85	111	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page